3 Computational thinking: Making a good plan

You will learn

- how to plan a program
- what an algorithm is
- what happens when you run a program.

In this unit you will learn to make and use a plan. A plan sets out the steps you have to take to solve a problem. Good planning can help you to achieve your goals.



Class activity

Another name for the plan for a task is an algorithm.

- What is your favourite food? Do you know how to make that food? Write down the actions in the right order.
- What is your favourite computer game? Write down what happens in the game. Put the actions in the right order.

These are examples of algorithms. Make an algorithm into a poster for the classroom wall.



Learning outcomes: Say what an algorithm is and what running a program means

algorithm
depends on
run a program
command

Talk about...

Share the recipe plans and game plans you made. Work together to make the plans even better. Can you add more actions? Can you add pictures?



Did you know?

On internet sites, such as YouTube, you can watch videos of people cooking their favourite foods.



3.1 What actions to take

In this lesson

You will learn:

- how to make a plan by choosing necessary actions.

Spiral back

Last year you used a program made of blocks. Each block stood for one action. Now you will learn to plan a program by putting actions in the right order.

The story of the donkey's hat

There was once an old lady who felt a little sad.



She made a hat for her donkey.

The old lady no longer felt sad.

Her grandson saw the donkey's hat. He asked her,
"How did you do it?"



Activity



Here are some actions that the old lady did that day:

- Peel the potatoes
- Sew flowers on the hat
- Find a straw hat
- Pick flowers
- Drink a glass of water
- Tidy the room
- Put the hat on the donkey's ears
- Make holes in the hat

Which actions are **necessary** to make the hat?
Necessary means actions that must happen.
'Tidy the room' is not necessary to make a hat.

Think again

Write down the things the old lady did. Only write the ones that are necessary to make the hat.

Extra challenge

Write the story of the donkey's hat in your own words. Draw a picture to go with the story.

3.2 The right order

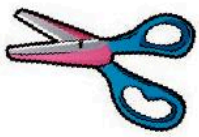
In this lesson

You will learn:

- how to make a plan by putting actions in the right order.

In the last lesson you picked necessary actions to make the donkey's hat. Here are some necessary actions.

Make holes in the hat



Sew flowers on the hat



Put the hat on the donkey's ears



Pick flowers



Find a hat



Now you will put the actions in the right order.

The donkey's hat fits on the donkey's ears. But first you have to make holes, so the ears can fit.

Putting the hat on the donkey **depends** on holes for ears.

One action uses the results of the other action. That's what 'depends on' means.



So, we can tell that

Make holes in the hat

must come before

Put the hat on the donkey's ears

The right order is called the **sequence** of actions.



Activity

Here are two actions:



Sew flowers on the hat



Pick some flowers

Which one has to come first? Say how you know.

Think again

Read all the actions on this page. Write the actions in the right sequence.



Extra challenge

Sometimes two actions are **independent**. That means they do not affect each other. It does not matter which comes first. What two actions in this story are independent? Say how you know.